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Report on the Examination

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REPORT ON THE EXAMINATION: INTERNATIONAL – JANUARY 2026

QUESTION A1.1

This question was well answered by most students. A number of students incorrectly suggested the missing particles would be an anti-neutrino. Some did not write 0 for the atomic and mass number and instead left the lines blank. It should be noted that in particle equation there is a need to explicitly give an answer if an answer line is given, even when the answer is zero.

QUESTION A1.2

This was well answered with the majority of students being able to correctly calculate the number of nuclei and the activity, giving the correct unit. A small number only gave their final answer to one significant figure (SF) and this was not accepted. In general more than 1 SF is required for calculations.

QUESTION A1.3

The majority of students got full marks for this question. However, some only gained 2 marks by calculating the reaming mass and not then calculating the mass that had decayed. Students managed to convert between the number of nuclei and mass well, with many recognising that the exponential calculation could be done directly with mass.

QUESTION A1.4

This proved to be a challenging question with most students stating that daughter nucleus later decayed via gamma emission, despite the question stating that the product was stable. In general only the higher scoring students thought about positron-electron annihilation to give the correct answer.

QUESTION A2.1

This question was well answered, although the perpendicular aspect was often left out. A small number of students gave a definition of magnetic flux density, rather than the Tesla, leaving out the specific aspect of 1m and 1A and answered instead more generally with *per unit length* or *per unit current*.

QUESTION A2.2

Many students correctly labelled the direction of the force and it was clear that Fleming's Left Hand Rule was understood well. Similarly, many gave good explanations for the direction of the force on the wire, with good reference made to Newton's Third Law. Most students did not state the direction of the current to give a complete answer, which could be done either by stating the direction or by clearly labelling the current direction on Figure 1.

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QUESTION A2.3

This was very well answered with only a small number not correctly converting units.

QUESTION A2.4

This was one of the most difficult questions on the paper. A number of students seemed to appreciate the idea that the vertical section would cause a difference but struggled to explain it clearly to get both marks.

QUESTION A3.1

Most students seemed to know that the field lines ought to be radial and symmetrical but there was a lot of carelessly drawn diagrams which did not then get the mark – either because of a clear lack of symmetry or the field lines not appearing to come from the centre. Students should use rulers to draw straight lines on questions like this.

QUESTION A3.2

This was answered well with most students knowing what to do to get the answer. A small number did not show their substitutions carefully enough. Working is required for full marks when the command word *determine* is used. In any case, it is good practice for students to show full working.

QUESTION A3.3

As this question was a *show that* the working needed to be clear and the answer must be given to more significant figures than the value shown in the question, in this case the answer had to be given to at least 2 significant figures. A number of students used equations with *k* as an electric force constant, rather than $\frac{1}{4\pi\varepsilon_0}$ as given in the specification and the data and formulae booklet. As this is a *show that* question students must use the version given in the data and formulae booklet, so this was not accepted for full marks. It is advised that centres teach the equation as given in the specification for electric fields. A small number of students confused their *k* with the Boltzmann constant in their substitution, which gives another reason why if is not advisable for students to use this version of the equation as k is not given in the data and formulae booklet.

QUESTION A3.4

This question differentiated well, with many good answers using a range of methods to calculate the kinetic energy required correctly. A few students took the analogy of gravitational orbits and applied the idea that the kinetic energy is half of the potential energy in a bound orbiting system. Some added the potential energy to the kinetic energy, rather than calculating the difference.

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It was impressive that most students were able to use electric fields and forces for an orbiting problem comfortably.

QUESTION A4.1

The vast majority of students answered this correctly

QUESTION A4.2

This question was also answered correctly by the majority of students with most scoring both marks. To score the first mark students needed to *use* either $\boldsymbol{C = Q/V}$ or $\boldsymbol{V = IR}$; this requires either a correct algebraic manipulation or a correct substitution of values to get the mark. A power of ten error in ignoring or not correctly taking into account the $\boldsymbol{\mu}$ was not penalised in the first mark – but this was a common reason for full marks not being scored along with algebraic errors in using the two equations together.

QUESTION A4.3

This was well answered by many students and nearly all managed to explain some aspects of the variation of intensity with light correctly. The majority of students scored at least half of the marks available, mostly by explaining that the rise in intensity was due to a rise in temperature, and that the decrease in intensity was due to cooling. Only a few student offered a convincing explanation for the temperature rise. A large number of students did recognise that the current would decrease with time due to the capacitor discharge.

QUESTION A4.4

Most students seemed unsure how to answer this question. Many gave a theoretical explanation of why the intensity should be exponential, perhaps missing the importance of the *determine* in the question. Those that did recognise the requirements to use Figure 6 to get data were mostly able to use data analysis to explain that the decrease was not exponential in nature. Correct approaches checked for consistent half-life or the consistency of the ratio of decrease at fixed time intervals.

QUESTION A4.5

About half of the students answered this correctly. Many realised that the resistance decreased but failed to link that to the discharge time. A significant minority of students made the mistake of thinking that the addition of the second lamp in parallel would increase the resistance of the circuit, sometimes then arguing that the student in the question is correct.

QUESTION A5.1

About half of the students answered this correctly. A number stated that the magnitudes were equal, without going on to explain that they were in opposite directions. A few stated that the fields were *equal and opposite* without explaining this in terms of magnitude and direction, this was required for the mark.

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QUESTION A5.2

About three-quarters of the students scored full marks on this question. Marks were lost most commonly due to a miscalculation or a mistake in the algebra, after correctly equating the gravitational field strength equations.

QUESTION A5.3

Fewer than half of the students answered this correctly. A number of students gave the definition of gravitational potential correctly but this in itself does not answer the question. Others explained that the gravitational potential is zero at infinity, some adding that the point P was not at infinity, but this was not credited as an explanation that answered the question.

QUESTION A5.4

This question differentiated well, with better answers recognising the need to find the gradient of the graph and to draw the tangent on the graph to do so. A number of students attempted to answer the question by using the equation for gravitational field strength, using the mass of the Earth. This method was not rewarded however, as it assumes that the equation for the gravitational field strength around a point mass could be used, when in this case the graph is showing the field due to both the Earth and the moon. A small number of students attempted to divide the value of potential at $\mathbf{2.0 \times 10^{8}}$ by this distance, rather than using the gradient, this approach was not credited.

A significant number of students gave an answer to 1 sf only, which did not get full marks. This happened across the paper, but was particularly noticeable on this question.

QUESTION A5.5

The majority of students calculated the energy required to move from T to S, using the difference in potential between T and S, not recognising that more energy would be required to move the object to point P first. One mark was given, however, for students that made this mistake. A small number of students calculated the energy required to completely free the object, by just multiplying the potential at T by its mass – this received no credit.

QUESTION A6.1

This was answered correctly by the vast majority of students.

QUESTION A6.2

Most students scored 1 mark here, but less than half scored full marks. Mostly correct used the SHM cosine equation to find ω, but a common mistake was to substitute $\mathbf{-1.8}$ cm for the displacement rather than $\mathbf{-1.8}$ cm.

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QUESTION A6.3

Most students answered this correctly. A small but noticeable number rounding their intermediate values and did not gain the final mark. Students should not round their answers until the final answer, using either 1 or 2 more SF than the final answer for intermediate values or using the full number from their calculator.

QUESTION A6.4

Most students correctly identified the time where maximum and zero kinetic energy occurred. Most lost the second mark by not drawing a sinusoidal variation, most commonly drawing cusps instead at T/2 and T. Some correctly drew the sinusoidal variation at T/2 and T, but did not start or finish the graph sinusoidally at 0 or 3T/2. Some students sketched the maxima kinetic energy decreasing and lost the second mark – either by careless sketching or by thinking that damping was taking place. The acceleration-time graph in Figure 11 showed no damping.

QUESTION A7.1

Most students answered this correctly. More would have got the mark were it not for careless drawing of the arrow. There were a lot of arrows drawn that did not touch the coin, or drawn so that it was not clearly aiming towards the centre of the plate. It is recommended that students use rulers when drawing vectors, as in question 03.1.

QUESTION A7.2

The majority of students gained full marks on this question. Students understood the question well, and how to apply the equations. The most common mistake was to use the circumference as the radius in the circular motion equations.

QUESTION A7.3

The vast majority of students recognised that the second angular velocity would be greater and many scored the second mark for explaining this well. Some did not mention the fact that the Force and the mass were constant. Some students successfully used an algebraic approach showing that $\boldsymbol{\omega^2 = 0.85g/r}$ which was an accepted and effective way of answering the question.

QUESTION A8.1

Readings were taken correctly from the graph but a number of students wanted to try to apply these values to BAN or to BAN/t rather than calculating the change of flux density over the time period.

QUESTION A8.2

Many students correctly stated that the frequency of the emf would decrease, although not all of these then explained why with reference to the frequency and time period. The majority explained that the amplitude decreased and most did explain this in terms of the change of flux. However, some only mentioned flux without mentioning the change. A significant number of students explained that the decreased frequency

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would also decrease the peak induced emf. Some students answered utilising the equation for peak emf = BANω, which was accepted.

SECTION B

Most of the multiple choice questions were answered well, many by over 75% of the students.

Questions which were found to be harder were : 15, 19 and 21. In these cases most students got the questions correct, but in smaller amounts than above.

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